

7 Decision analysis	7.1	Use, construct and interpret decision trees.	Students should be familiar with the terms <i>decision nodes</i> , <i>chance nodes</i> and <i>pay-offs</i> .
	7.2	Use of expected monetary values (EMVs) and utility to compare alternative courses of action.	

Glossary for Decision Mathematics 2

Transportation problems

In the **north-west corner method**, the upper left-hand cell is considered first and as many units as possible sent by this route.

The **stepping stone method** is an iterative procedure for moving from an initial feasible solution to an optimal solution.

Degeneracy occurs in a transportation problem, with m rows and n columns, when the number of occupied cells is less than $(m + n - 1)$.

In the transportation problem:

The **shadow costs** R_i , for the i th row, and K_j , for the j th column, are obtained by solving $R_i + K_j = C_{ij}$ for **occupied cells**, taking $R_1 = 0$ arbitrarily.

The **improvement index** I_{ij} for an **unoccupied cell** is defined by $I_{ij} = C_{ij} - R_i - K_j$.

Flows in networks

A **cut**, in a network with source S and sink T , is a set of arcs (edges) whose removal separates the network into two parts X and Y , where X contains at least S and Y contains at least T . The **capacity of a cut** is the sum of the capacities of those arcs in the cut which are directed from X to Y .

If a problem contains both **upper** and **lower capacities** then the capacity of the cut is the sum of the upper capacities for arcs that cross the cut in the direction from S to T minus the sum of the lower capacities for arcs that cross the cut in the direction from T to S .

If a vertex has a **restricted capacity** then this can be replaced by two unrestricted vertices connected by an edge of the relevant capacity.

If a network has several sources $S_1, S_2, \dots$, then these can be connected to a single **supersource** S . The capacity of the edge joining S to S_1 is the sum of the capacities of the edges leaving S_1 .

If a network has several sinks $T_1, T_2, \dots$, then these can be connected to a **supersink** T . The capacity of the edge joining T_1 to T is the sum of the capacities of the edges entering T_1 .

Dynamic programming

Bellman's principle for dynamic programming is 'Any part of an optimal path is optimal.'

The **minimax route** is the one in which the maximum length of the arcs used is as small as possible.

The **maximin route** is the one in which the minimum length of the arcs used is as large as possible.

Game theory

A **two-person game** is one in which there are exactly two players.

A **zero-sum** game is one in which the sum of the losses for one player is equal to the sum of the gains for the other player.

Decision analysis

At a **decision node**, the user must choose the most favourable outcome.

At a **chance node**, the Expected Monetary Value is calculated.

The **end nodes** show the profit for each outcome.